

TECHNICAL ENGINEERING PORTFOLIO

WEIYI HUANG

Electrical Engineering | Avionics | Embedded Hardware

Anduril Internship Application

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01 | TECHNICAL PROFILE

ENGINEERING SUMMARY

Engineer focused on avionics integration, high-voltage UAS architectures, and high-speed embedded hardware. Led electrical and embedded integration at an acquired UAS startup, owning hardware from architecture through bring-up, failure analysis, and flight validation. Current Junior studying Electrical and Computer Engineering and Aerospace Engineering at Cornell University.

SELECTED TECHNICAL EVIDENCE

AREA	PROOF POINT
Embedded compute carrier	Six-layer CM5 carrier integrating USB, Ethernet, power, and aircraft I/O; architecture revised after requirements and integration testing; deployed with 200+ flight-hours.
High-voltage UAS power	~80 kW (peak) aircraft power architecture; ESC failure investigation using bench testing, oscilloscope measurements, component analysis, and fault-tree reasoning.
Aircraft networks & integration	CAN/UAV bus SI mitigation; analysis of CAN bus using bench testing, Kvaser, and FC packet measurements; increased message reception rate by ~60%.

CORE TOOLKIT

Systems & Architecture: Full-lifecycle EE hardware ownership / first-principles electrical architecture / Design for Manufacturing (DFM) & interface consolidation

Power & High-Voltage: High-voltage bus architecture for UAS systems / HV protection & fault-isolation design / high-voltage wire harness topology & busbar design

High-Speed & Embedded: Multi-layer high-speed PCB design & EMI mitigation / CAN bus architecture & bandwidth headroom optimization / low-level embedded C firmware debugging (STM32/SWD)

Test & Validation: Rapid root-cause failure analysis & hardware triage / automated test stand engineering & dynamic sensor characterization / avionics systems bring-up & flight-test validation

02 | EMBEDDED COMPUTE CARRIER FOR AUTONOMOUS UAS

ROLE: Lead EE | TIMELINE: 6 months | STATUS: 200+ flight-hours

Requirements: Initial architecture -- USB 3.0 + 1000BASE-T

Revised architecture -- USB 2.0 payload + USB 3.0 camera + 100BASE-TX radio interfaces

Constraints: ~3.5 inches width limit, constrained connector placement, vibration-resistant mounting method

What I owned: Schematic + layout, bring-up, debug, revisions, validation.

KEY ENGINEERING DECISION

Initial architecture: Six-layer, controlled-impedance carrier in a ~3.5" x 1.5" footprint: Raspberry Pi CM5 + USB 3.0 + 1000BASE-T + ~50 W power subsystem, with Ethernet and USB brought out through vibration-tolerant locking interfaces. Link speeds were sized against the anticipated bandwidth of the IP radio and CV sensors.

Integration findings: The selected IP radio ultimately supported only 100BASE-TX over two-pair wiring, so the two unused Gigabit pairs contributed no traffic while adding a resonant coupling path alongside the active pairs. Sustained USB 3.0 throughput proved equally unusable — processing 5 Gbps from the CV sensor exceeded the CM5's available thermal envelope. NDAA sourcing constraints also favored multiple discrete CV sensors over integrated units, making USB port count more valuable than peak per-port bandwidth.

Revised architecture: Moving from 1000BASE-T to 100BASE-TX removed two differential pairs and reduced the Ethernet magnetics footprint. One USB 3.0 port was retained for the plausible class of sensors that could actually use it, and two USB 2.0 ports were added to prioritize port quantity and sourcing flexibility. The net trade converted unused bandwidth into routing margin, board area, signal isolation, and additional I/O.

PCB HIGHLIGHTS

- Ground plane under and around high-speed differential traces.
- Ground stitching concentrated around high-speed differential traces and return-path discontinuities, spaced for reducing high-frequency return loops and limit radiated EMI.
- 50 Ω single trace impedance calculated through Saturn PCB Design Toolkit.
- 90 Ω /100 Ω differential pair impedance calculated through Saturn PCB Design Toolkit.
 - 3W/5W inter-pair gap between unrelated differential pairs to minimize coupling.
 - Inter- and intra-pair length matching for sensitive, high-speed signals.
- Power trace width calculations for voltage sag characteristics.

Validation: Each revised interface was validated at bring-up against its intended peripheral — USB 2.0 payload ports enumerated at High Speed (480 Mbps), the USB 3.0 camera port sustained SuperSpeed throughput (5 Gbps), the radio interface linked at 100BASE-TX, and both the CM5 and USB PD rails held their voltage thresholds with minimal sag under 100% CPU stress and scoped transient bench loading. See Table 1 below for full testing methods.

Result: The revised architecture is the actively deployed configuration, with 200+ flight-hours.

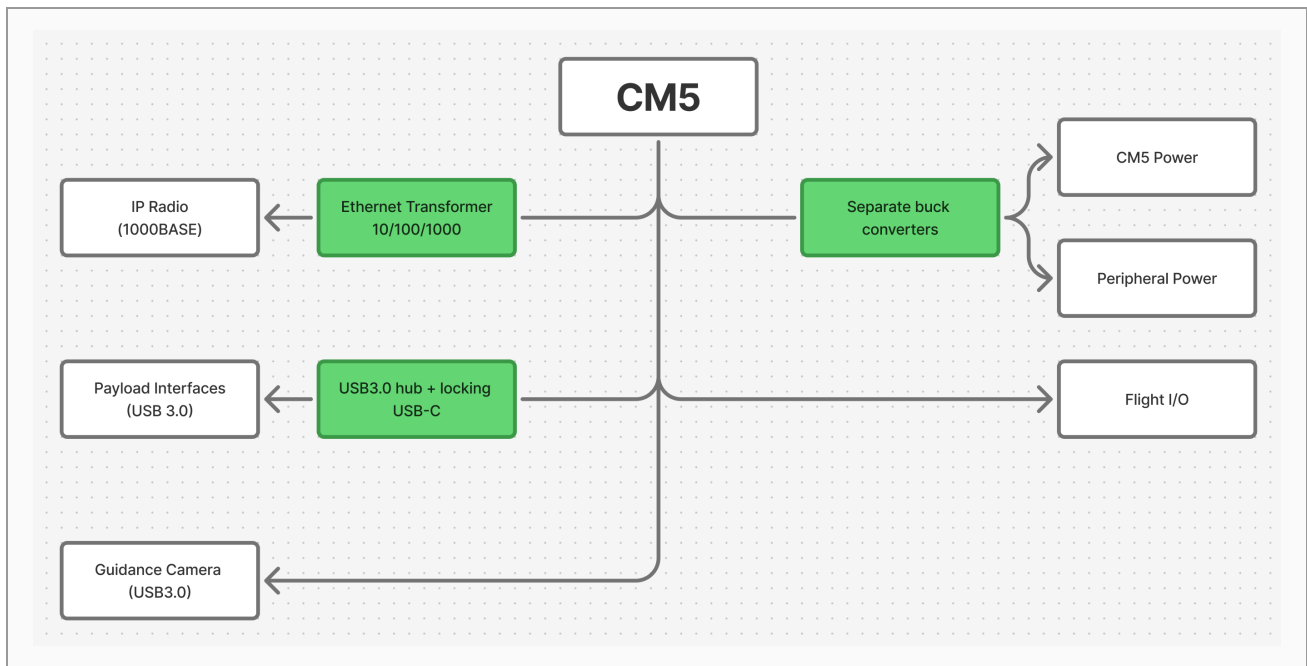


Figure 1. Generalized compute-carrier architecture BEFORE integration and tradeoff revisions.

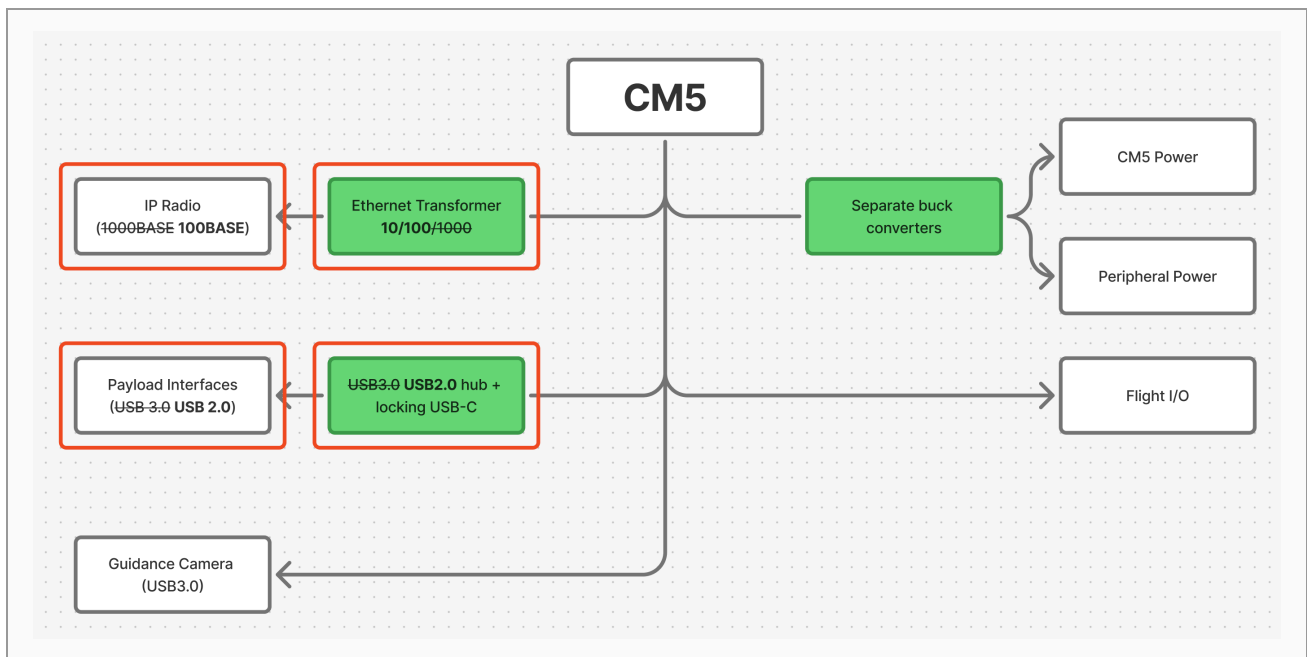


Figure 2. Generalized compute-carrier architecture AFTER integration and tradeoff revisions. Revisions are outlined in red.

REQUIREMENT / RISK	VALIDATION METHOD	EVIDENCE	RESULT / STATUS
USB 2.0 – Payload Interfaces	dmesg (kernel messages upon connection) sudo dd (throughput test) Specific Python script	Validation with intended USB peripheral	Enumerated at USB 2.0 High Speed (480 Mbps); validated with intended peripheral
USB 3.0 – Guidance Camera	dmesg (kernel messages upon connection) sudo dd (throughput test) Specific Python script	Validation with intended guidance camera	Enumerated at USB 2.0 High Speed (480 Mbps) in bootloader; operating throughput at USB 3.0 SuperSpeed (5 Gbps) validated with intended peripheral
Ethernet – IP Radio	dmesg (kernel messages upon connection)	Validation with IP radio and link bandwidth	100BASE-TX
Power – CM5	dmesg (verify no undervoltage warnings) stress -c (CPU stress load) Bench load (through test pads) + scoped rail measurements	Stress tested CM5 to 100% CPU utilization Measured transient events with bench load	Maintained operation and voltage threshold with minimal voltage sag
Power – Peripheral	dmesg (verify no undervoltage warnings) stress -c (CPU stress load) Bench load (through test pads) + scoped rail measurements	Drew steady-state and transient load from intended port	Maintained voltage threshold with minimal voltage sag

Table 1. Bring-up and validation matrix for the revised compute-carrier architecture.

03 | HIGH-VOLTAGE UAS / ESC FAILURE INVESTIGATION

PRIMARY STORY: fault isolation, instrumentation, risk containment, and retest

ROLE: Lead Investigator and Lead EE | TIMELINE: 72 hours | STATUS: Repaired and deployed

Failure signature: During pre-flight commissioning of a 100 V UAS platform, two out of eight ESCs suffered catastrophic thermal events while idling at low current (<100 mA) with control signals disconnected. Scorching was present on the adjacent carbon-fiber structure around the ESC mounting points. The platform was grounded five days ahead of a committed demo-flight shipment.

System context: ~100 V HV bus, ~80 kW peak power, eight motor controllers on a shared harness, long cable runs, an electrically conductive carbon-fiber airframe, and a high-vibration flight environment.

What I owned: Complete redesign and integration of HV wire harness within 72 hours.

INVESTIGATION LOGIC

Path: unreproduced thermal failure at idle → physical evidence and teardown → ranked risk set → 72-hour HV harness re-architecture → retest and flight clearance.

- **Step 1 — Reproduce:** Bench testing at full bus voltage did not reproduce the event, and measured bus behavior showed no repeatable idle-condition overvoltage. With a non-reproducible fault and a committed shipment five days out, I shifted from root-cause confirmation to physical evidence, teardown, and a containment-first plan.
- **Step 2 — Teardown:** Teardown found capacitors rated at 100 V operating on a 100 V bus with minimal clearance to the chassis shell. The scorch pattern on the surrounding structure appeared consistent with an energized chassis establishing an HV path through the low-resistance carbon fiber.
- **Step 3 — Theorize:** Collected evidence narrowed the cause to two credible mechanisms: an internal capacitor failure from inadequate voltage rating, and an internal short from a QC escape. Scorch pattern suggests a HV path as the plausible mechanism for failure propagation from the first ESC to the second.
- **Step 4 — Rank risks:** Identified zero-margin capacitor voltage derating, insufficient chassis clearance, and transient susceptibility as the dominant risks from available evidence.
- **Step 5 — Contain and rebuild:** Led a team of two through a 72-hour containment redesign of the HV harness, including:
 - Properly derated DC-link bulk capacitance at each ESC, offloading transient voltage stress from the underrated internal capacitors
 - Fast-acting fusing at the battery and at each ESC branch for fault isolation
 - A removable precharge/discharge circuit to limit capacitive inrush
 - Written HV energization, discharge, and emergency protocols.

Validation: Validated the redesign through 100 V static soak testing and motor sweeps across the full throttle envelope on all eight channels, with pass criteria of no thermal event and stable operation at each throttle setpoint.

Result: Platform cleared for flight within 72 hours (two days ahead of the shipment deadline) with no recurrence of the thermal event in demo-flight. Branch-level fusing and isolation substantially reduced the risk of multi-ESC fault propagation.

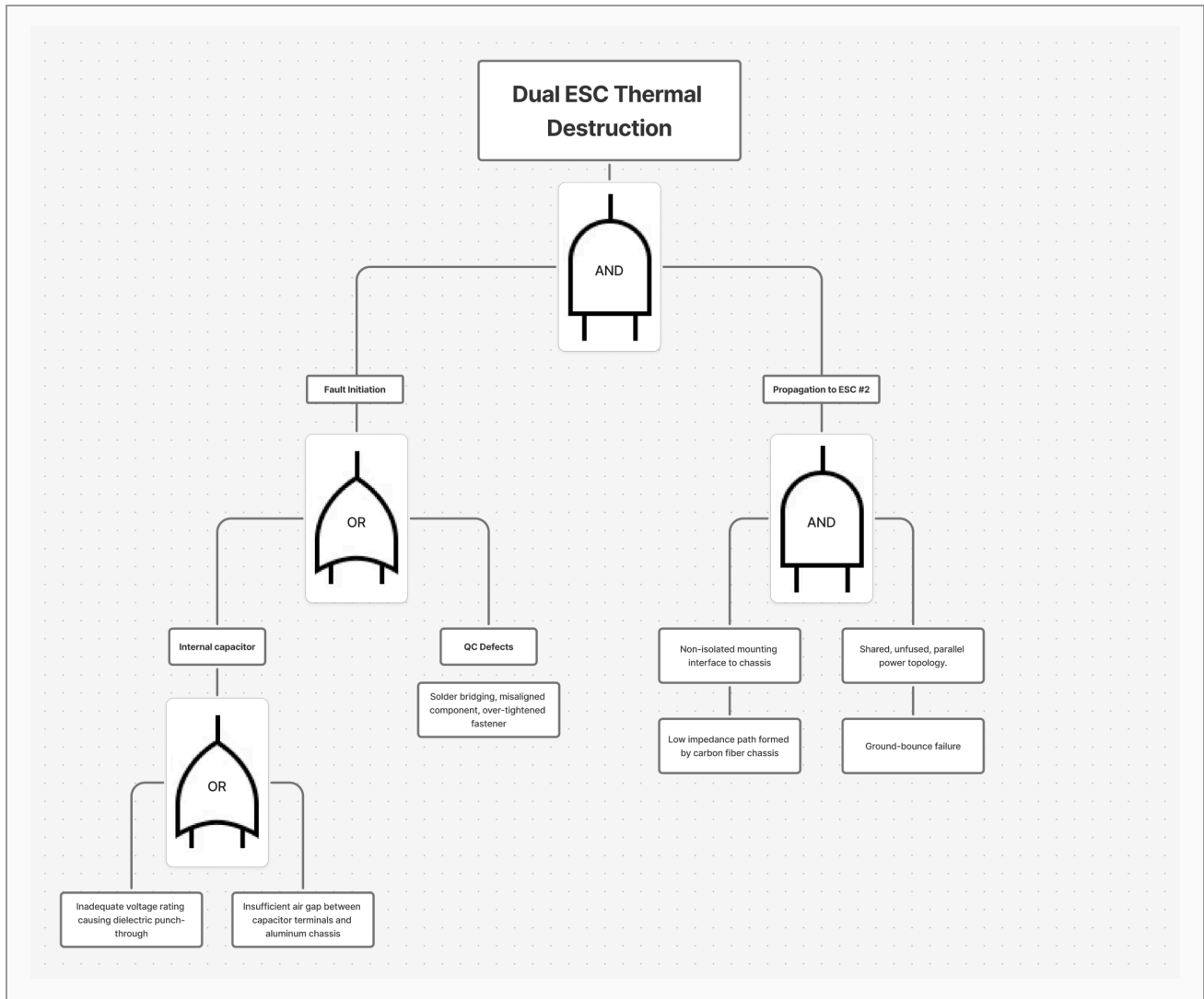


Figure 3. Failure-analysis flow for ESC failure. Recreated conceptually.

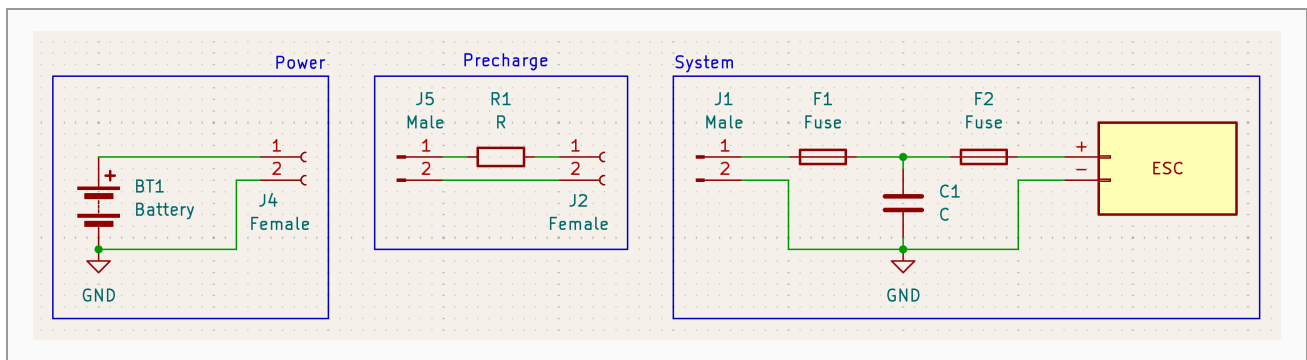


Figure 4. Generalized wire harness diagram with capacitor and branch fusing implemented. After precharging is complete, power circuitry is plugged directly into system with precharge eliminated.

04 | AIRCRAFT ELECTRICAL & CAN INTEGRATION

SECONDARY STORY: systems engineering, interfaces, harnesses, and flight-test closure

ROLE: Lead Investigator and Lead EE | TIMELINE: 2 weeks | STATUS: Deployed and flight-tested

Failure signature: Flight logs showed intermittent ESC telemetry dropouts in flight, with propellers attached and the aircraft drawing high current. The same symptoms were not reproducible on the ground with propellers removed at low current draw, which pointed away from a device-level fault and toward the shared CAN network itself. Inspection of the inherited wiring confirmed an incorrect bus topology.

System context: Telemetry dropouts reduced propulsion-network robustness and created an unacceptable flight-control reliability risk.

What I owned: Complete re-architecture of the aircraft CAN network — topology, harness, and termination — from diagnosis through bench validation and flight-test closure.

DESIGN CONSIDERATIONS

- **Topology and termination:** The inherited network was wired as a star, the incorrect topology for using it as a shared bus between devices. I re-architected it as a linear bus topology with short device stubs and 120 Ω termination at both physical endpoints, allowing the bus to operate in its intended regime.
- **Stub reflections:** Trunk wires were consolidated so device stubs stay short relative to the trunk, limiting the reflections and discontinuities that accumulate over long airframe cable runs and repeated connector transitions. After new topology was validated through bench testing, wiring was consolidated into a PCB for improved signal integrity and consistency across aircraft.
- **EMI susceptibility:** The load-correlated failures suggested motor-drive noise was consuming the limited noise margin of the star network. The revised topology increased the network's noise margin.
- **Diagnostic access:** The revised harness implemented connectorized terminations for easy diagnostic access, allowing frame rates and error behavior to be captured on the bench and in-flight configuration.

Validation: Verified the revised topology on the bench with a Kvaser Leaf, confirming no packet loss, and in-flight configuration by inspecting flight-controller CAN RX packet rates.

Result: Under equivalent node and message configurations, CAN RX increased ~60% and also provided sufficient signal-integrity margin to operate reliably at the higher CAN baud rate. The bus held stable at the new baud rate with no overload events, which the legacy harness could not support given its signal integrity. The frame rate remained steady across the full flight period.

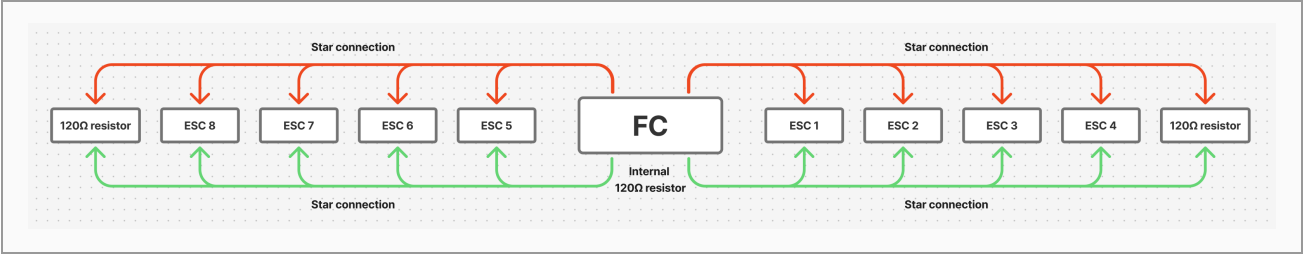


Figure 5. Initial aircraft CAN bus wiring topology. Recreated conceptually.

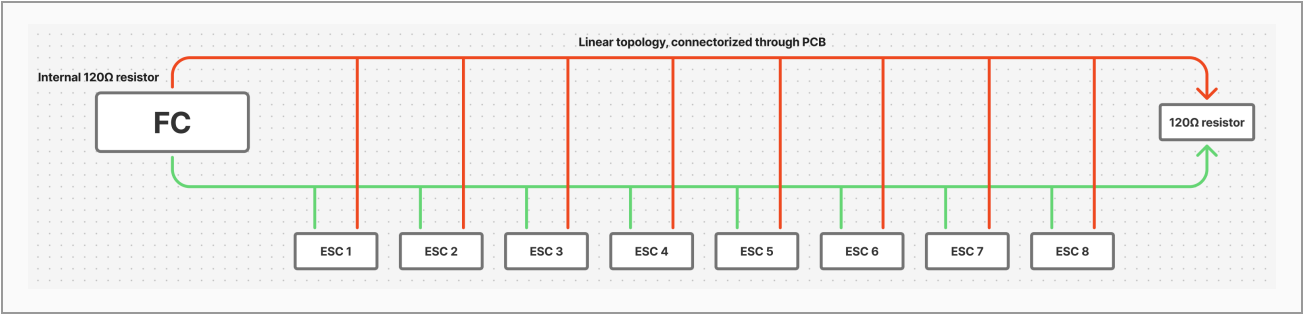


Figure 6. Revised aircraft CAN bus wiring topology. Recreated conceptually.